VA50FA2V-40

40x Optical Zoom PTZ Camera

Ambarella CVflow

QCVN 135:2024/BTTTT



The 40X Optical Zoom PTZ Camera is an AI camera product "Made in Vietnam" by MK Vision, providing a comprehensive surveillance solution that combines advanced image processing technology with Edge AI (AI on Edge). The camera offers flexible monitoring capabilities with a wide 360° horizontal and 90° vertical field of view, combined with 40x optical zoom and 10x digitalzoom,enablingdetailedtrackingoverlongdistances.

The PTZ camera integrates HDR Engine Gen3 to optimize image quality in backlit conditions and uses Ambarella CVflow to support intelligent video content analysis with edge AI processing. The device is designed to meet IK10 impact resistance and IP67 dust and water protection standards, and fully complies with ONVIF Profiles S/G/T/M as well as the information security standard. The 360° PTZ camera ensures outstanding performance, durability, and a high level of security for large-scale surveillance projects.

- **40X optical zoom, 16X digital zoom**
- **Viewing range: 360° horizontal, 90° vertical**
- **Extended long-range infrared distance**
- **Impact resistance IK10**
- **HDR Engine Gen3**
- **Video content analysis**
- **Dust and water protection IP67**

Headquarters: The Vista, No. 4, Lane 15
Duy Tan Street,Cau Giay District, Hanoi

Website: http://mkvision.com
Email: contact@mkvision.com

Phone: (+84 24) 7100 6785
(+84 24) 3813 4646

Specification

Configuration	
SoC	Ambarella CV25
Processor	Quad-core ARM® Cortex® A53 1GHz
ROM memory	NAND Flash
Memory	1GB DDR4
Processing performance	CVflow® Computer Vision Engine @588MHz DNN Accelerated Engine
Smart features	AI Video Motion Detection, AI People Detection, AI Vehicle Detection, AI Face Detection, AI Face Recognition, AI License Plate Recognition, ...
Camera	
Sensor	Ambarella CV25
FOV	Quad-core ARM® Cortex® A53 1GHz
Lens	NAND Flash
Detection range 25 ppm	1GB DDR4
Observation range 62.5 ppm	CVflow® Computer Vision Engine @588MHz DNN Accelerated Engine
Recognition range 125ppm	AI Video Motion Detection, AI People Detection, AI Vehicle Detection, AI Face Detection, AI Face Recognition, AI License Plate Recognition, ...
Identification range 250 ppm	5,6m(W) - 24m(T)
Night mode	Automatic IR-cut filter Automatic switching
Minimum lighting	Infrared mode: 0 lux Infrared: ≥ 50m
Shutter speed	1 ~ 1/10.000 s
PTZ mode	Maximum horizontal rotation angle: 0°–360°, vertical rotation angle: -20° to 100°. Horizontal rotation speed: 0.1°–90°/s, vertical rotation speed: 0.1°–55°/s
Video	
Resolution	5 Megapixels (2592H x 1944V)
Compression standard	H.265 / H.264 / Motion JPEG
Frame rate	5MP @30fps
Signal-to-Noise Ratio (SNR)	≥ 50dB
Exposure balance	HDR Engine Gen3, WDR 120dB
Adjust the image	Automatic White Balance (AWB), Backlight Compensation (BLC), Automatic Gain Control (AGC), Auto Iris, Auto Focus
Connect	
Connection ports	RJ45, 10/100/1000Mbps
Supports protocols	ARP, PPPoE, IPv4/v6, ICMP, IGMP, QoS, TCP, UDP, DHCP, UPnP, SNMP, SMTP, RTP, RTSP, HTTP, HTTPS, FTP, NTP, DDNS
ONVIF®	Profile S, G, T, M
Certification	
ISO	ISO 9001:2015
Protection standard	IP67 - TCVN 4255:2025 (IEC 60529:2013) IK10 - TCVN 14227:2025 (IEC 62262:2021)
Electrical safety	TCVN 73126-1:2003 (IEC 60950-1:2001)
Information safety	NDAA, FIPS 140-2 Level 3
Overview	
Case	Metal
Power supply	24VAC; PoE IEEE 802.3bt, Type 3, Class 6
Storage	Support microSD / microSDHC / microSDXC
Operating conditions	-5 ~ +60°C/ humidity ≤ 95% RH
Storage conditions	-15 ~ +65°C/ humidity ≤ 95% RH
Dimensions	Ø 241x349 mm
Weight	5,5 kg
Supported languages	English, Vietnamese
Certificate of origin	Made in Vietnam

Headquarters: The Vista, No. 4, Lane 15
Duy Tan Street,Cau Giay District, Hanoi
Website: http://mkvision.com
Email: contact@mkvision.com
Phone: (+84 24) 7100 6785
(+84 24) 3813 4646